
ECON 312 — Intermediate Microeconomics

Lecture Notes — Ch. 3: Preferences

Fall 2026

Chapter 2 was all about what a consumer *could* choose — the budget constraint. Now we turn to *should*: which of the affordable bundles would the consumer actually choose?

→ Why “should”? We don’t yet know *what* the consumer will choose — but we *can* restrict *how* they will choose.

Outline

- ① Preferences vs. choices
- ② Three axioms: completeness, reflexivity, transitivity
- ③ Indifference curves, and four families of preferences
- ④ Well-behaved preferences: monotonicity & convexity
- ⑤ The marginal rate of substitution

0. Consumption Bundles

We will study consumers choosing **consumption bundles**.

→ A bundle is just a complete “group of things” the consumer does.

↔ i.e. what to do with a day — a plan for their entire life

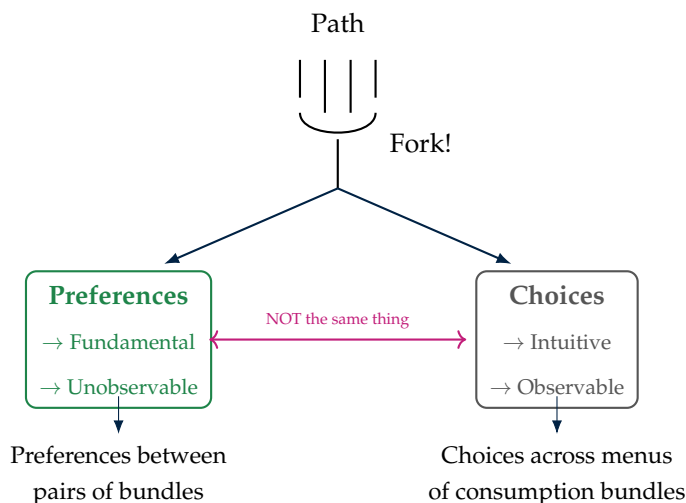
↔ eating an apple vs. an orange

* Anything *not* directly involved in the comparison is held constant (*ceteris paribus*).

We write a bundle as X , or in coordinates $(X_1, X_2), \dots$

2. Preferences vs. Choices

Goal: understand what a consumer does.



A **preference** is just a relation $\succsim$ over consumption bundles.

→ $X \succsim Y$ means “ X is at least as preferred as Y ” — i.e. *weakly preferred to* Y .

* Resembles the “greater than or equal to” symbol — in L^AT_EX it’s `\succsim`.

Strict Preference and Indifference

Similar to $\succsim$, we also define $\succ$ and $\sim$.

→ **Strict preference** ($\succ$): if $X \succsim Y$ and not $Y \succsim X$, then $X \succ Y$.

Q How should we think of $X \succ Y$ in terms of choices?

→ Y is never chosen → X is chosen instead.

→ **Indifference** ($\sim$): if $X \succsim Y$ and $Y \succsim X$, then $X \sim Y$ — X and Y are “as good” as each other.

3. Three Axioms

Q But is $\succsim$ just “ $\geq$ ” for consumption bundles?

No. Consider a consumer deciding between ice cream and cake:

ice cream $\not\succsim$ cake (NOT $\succsim$) and cake $\not\succsim$ ice cream.

This is *not* indifference — they don’t think ice cream is at least as preferable as cake, and don’t think cake is at least as preferable as ice cream.

* This is actually surprisingly true of humans. There are many things I just can’t compare.

→ But this would make analysis really tough, so we assume it away. Add an **axiom** — an unprovable thing you state is true.

A.1 — Completeness

For every X, Y : either $X \succsim Y$, or $Y \succsim X$, or both.

Is A.1 true? No, but it is close enough!

But is $\succsim$ reasonable now? What if I said:

cake $\not\succsim$ cake $\Rightarrow$ cake isn't at least as good as cake!

Does this make sense? No, not for preferences! So we add a second axiom.

A.2 — Reflexivity

For every X : $X \succsim X$.

Is A.2 true? I'd say yes...

But are we there yet? Are these “preferences”? What about *this* setup?

ice cream $\succsim$ cake, cake $\succsim$ donut, but ice cream $\not\succsim$ donut ?

That seems odd, eh? (Actually, my toddler has preferences like this sometimes.)

A.3 — Transitivity

If $X \succsim Y$ and $Y \succsim Z$ (we typically write this $X \succsim Y \succsim Z$), then $X \succsim Z$.

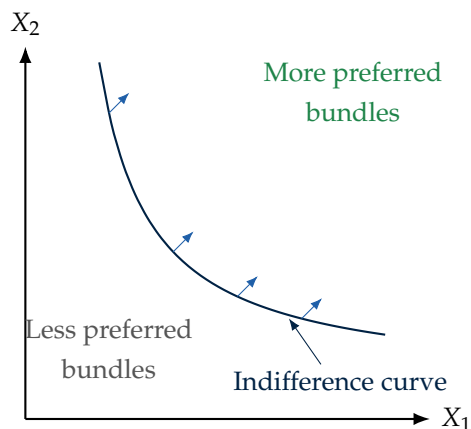
True? True enough! These three are good enough for us to work with.

* (Not 100% true — we actually need a few ‘technical’ assumptions too, but let's not worry about those.)

4. Indifference Curves

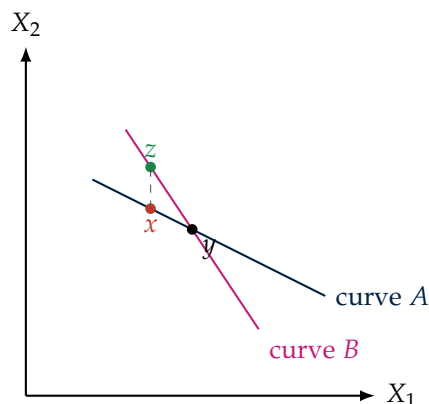
What can we do with these three assumptions? It turns out we can describe preferences over bundles with just one set of objects — **indifference curves**.

→ An indifference curve connects bundles Y with $X \sim Y$ on the X_1 – X_2 space.



Fact: indifference curves never cross each other!

Q Why is this a fact? What would it mean if they crossed?

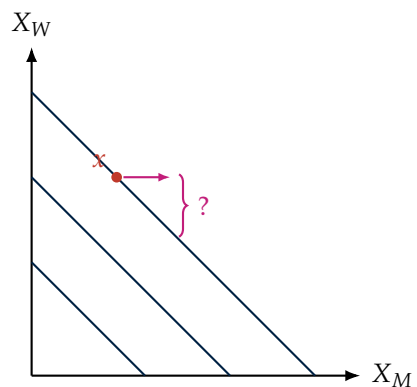


Suppose curves A and B crossed at y . Fix X_1 to the left of the crossing: z (on curve B) has strictly more X_2 than x (on curve A) at that same X_1 , so $x \not\sim z$ b/c not on same indifference curve. But $z \sim y$ (same curve B) and $y \sim x$ (same curve A), so transitivity gives $z \sim x$ — a contradiction!

Four Families of Preferences

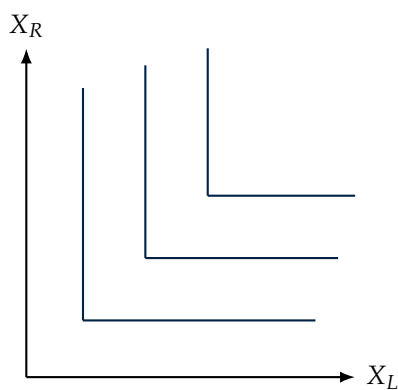
Let's try to draw some indifference curves to match some simple preferences.

1. Perfect Substitutes. Meijer water vs. Walmart water — only the total # of bottles matters.

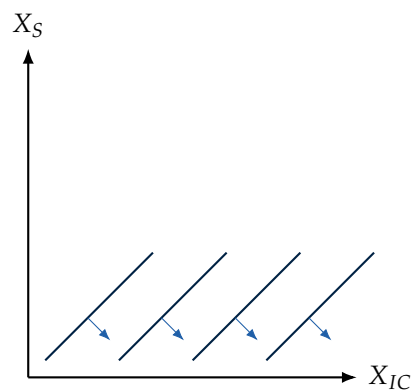


- (a) Put your pencil down somewhere (x).
- (b) Ask: “how would I change x to keep indifference?”
- (c) Repeat...

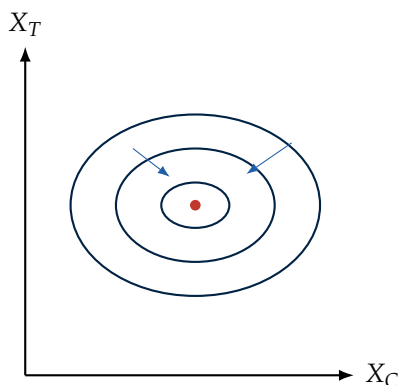
2. Perfect Complements. Left shoes and right shoes — only the total # of *pairs* matters.



3. Bads. Ice cream & toxic sludge — more ice cream $\Rightarrow$ good, more sludge $\Rightarrow$ bad.



4. Satiation. “Goldilocks” preferences — just the right amount, e.g. chocolate & tea.



We could go on... Now, indifference curves could take all sorts of shapes, and we need to restrict further to “well-behaved” preferences. That means adding **assumptions** — similar to axioms, but about the *shape* rather than logical coherence.

5. Well-Behaved Preferences

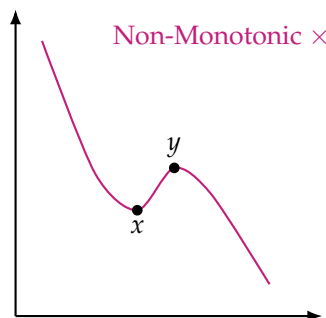
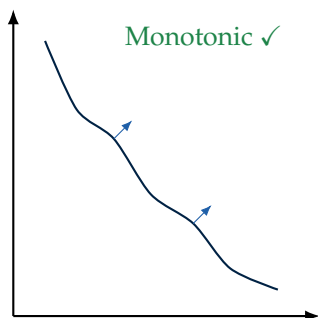
Assumption 1: Monotonicity

Assumption 1 — Monotonicity (“More is better”)

If $X_1 \geq Y_1$ & $X_2 \geq Y_2$, then $(X_1, X_2) \succeq (Y_1, Y_2)$.

If $\succeq$ is strict for any coordinate, then $(X_1, X_2) \succ (Y_1, Y_2)$.

Q Why is this reasonable? It allows for “free disposal” of extra goods.



An indifference curve for monotonic preferences always has a downward slope.

Q Why? If it sloped upward anywhere, the farther-up point would have to be preferred — monotonicity gives $y \succ x$, contradicting $x \sim y$.

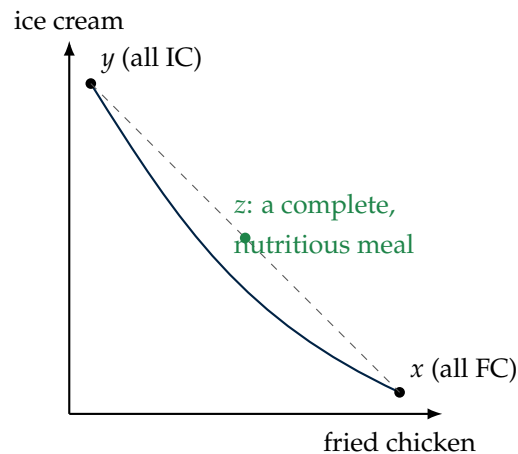
Assumption 2: Convexity

Assumption 2 — Convexity (“Averages are better”)

If $X \sim Y$, then $\frac{1}{2}X + \frac{1}{2}Y \succsim X$ (or Y).

(Any $t \in (0, 1)$: $tX + (1 - t)Y \succsim X$.)

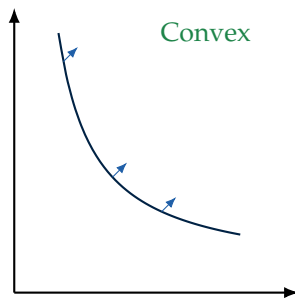
Q Why is this reasonable? Assume X and Y are bundles of ice cream and fried chicken:



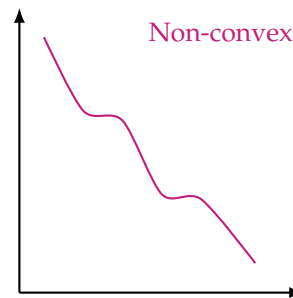
Note: we need $X \sim Y$! Otherwise, if $X = \$1$ million and $Y = \$0$, would we really want $\$0.5\text{m} \succsim \1m ?

Typically, goods have *some* complementarities, because consuming only one good isn't good.

→ I study problems where this doesn't hold.



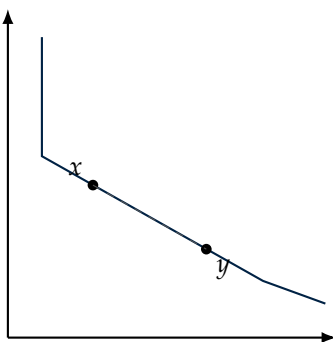
Convex



Non-convex

Q What restriction does this put on the shape? A non-decreasing (in absolute value) slope — the curve must bow toward the origin.

But how about this? Is this convex?



Yes, but it is different b/c the middle bundle is only *weakly* preferred to the extreme bundles. So here we make a distinction:

(Weak) Convexity

$$tX + (1-t)Y \succeq X \text{ for } t \in (0,1)$$

Strict Convexity

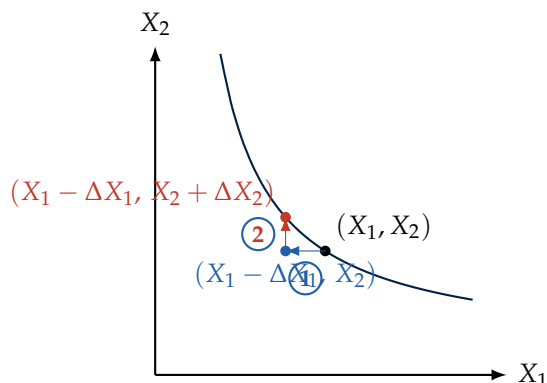
$$tX + (1-t)Y \succ X \text{ for } t \in (0,1)$$

↪ This has to do w/
smoothness of the
curve

6. The Marginal Rate of Substitution

Idea: how can we “swap” small amounts of X_1 and X_2 and still stay on the same indifference curve?

→ It is going to help us describe what the consumer chooses.



- ① Take a teeny-tiny bit of X_1 away... call this ΔX_1 .
- ② Add back in a teeny-tiny ΔX_2 . Lands on the OG indifference curve.

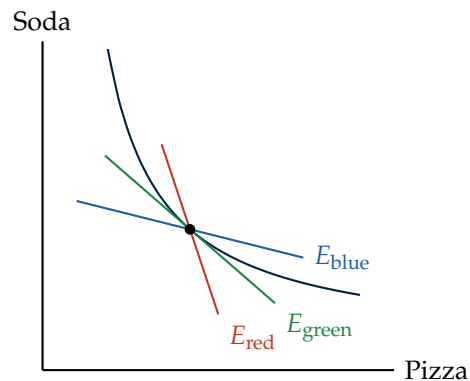
What $\frac{\Delta X_2}{\Delta X_1}$ is, is the slope of the indifference curve at (X_1, X_2) .

* Using calculus, we'd let ΔX_1 become infinitely small — dX_1 ...

We call $\frac{dX_2}{dX_1}$ the **Marginal Rate of Substitution**.

- **Substitution** → we are exchanging one good for another
- **Rate** → we are looking at a ratio
- **Marginal** → we are making a small change → Derivative!

Now think back to the consumer. Suppose we offered an exchange rate E (like 2 pizza per 1 soda)... When would they not use it? (That works both ways.)



Both E_{red} & E_{blue} cut through the indifference curve:

- Red would trade for more soda,
- Blue for more pizza.

Only at green do they *not* trade! → green is the MRS!

Note: MRS is always a negative #. Gaining one good means losing the other!

Two Last MRS Thoughts

- 1: When good X_2 is \$\$\$, the MRS is also the **marginal willingness to pay** → you are indifferent between keeping the money and spending it to buy good X_1 .
- 2: MRS changes based on X_1 & X_2 . For strictly convex and monotonic preferences, MRS *decreases* (in absolute value) as X_1 increases.
 - ↪ A “diminishing marginal rate of substitution.”